

1. Angles in spontaneous disintegration: In the laboratory system, find: (a) the relation between the angles θ_1 and θ_2 after a spontaneous disintegration of a particle into two particles, and (b) the angular distribution of the resulting particles.

a) $\vec{v}_i = \vec{v}_{i0} + \vec{V}$

$$x = \frac{V}{v_{i0}}$$



$$\tan \theta_1 = \frac{v_{i0} \sin \theta}{V + v_{i0} \cos \theta} = \frac{\sin \theta}{x + \cos \theta}$$

$$\begin{aligned} \tan \theta_2 &= \frac{v_{20} \sin \theta}{V - v_{i0} \cos \theta} \\ &= \frac{\sin \theta}{\frac{m_2 x}{m_1} - \cos \theta} \end{aligned}$$

$$\frac{m_1 v_{i0}}{m_2} = v_{20}$$

$$\Rightarrow x \tan \theta_1 + \cos \theta \tan \theta_1 = \sin \theta$$

$$\frac{m_2}{m_1} x \tan \theta_2 - \cos \theta \tan \theta_2 = \sin \theta$$

$$\Rightarrow x \tan \theta_1 + \tan \theta_1 \cos \theta = \frac{m_2}{m_1} x \tan \theta_2 - \tan \theta_2 \cos \theta$$

$$\cos \theta = \frac{\left(\frac{m_2}{m_1} \tan \theta_2 - \tan \theta_1 \right) x}{\tan \theta_1 + \tan \theta_2}$$

— (1)

Also,
$$\frac{\sin \theta - x \tan \theta_1}{\tan \theta_1} = \frac{\sin \theta - \frac{m_2}{m_1} x \tan \theta_2}{- \tan \theta_2}$$

$$\begin{aligned} - \tan \theta_2 \sin \theta + x \tan \theta_1 \tan \theta_2 &= \tan \theta_1 \sin \theta - \frac{m_2}{m_1} x \tan \theta_1 \tan \theta_2 \\ x \tan \theta_1 \tan \theta_2 \left(1 + \frac{m_2}{m_1} \right) &= \sin \theta \end{aligned}$$

$$\frac{\sin \theta}{\tan \theta_1 + \tan \theta_2}$$

— (2)

From 1 and 2,

$$x^2 \tan^2 \theta_1 \tan^2 \theta_2 \left(1 + \frac{m_2}{m_1}\right)^2 + x^2 \left(\frac{m_2}{m_1} \tan \theta_2 - \tan \theta_1\right)^2 = (\tan \theta_1 + \tan \theta_2)^2$$

with $x = \frac{V}{v_{10}}$

(make it more symmetric)

b) $\tan \theta_1 = \frac{\sin \theta}{x + \cos \theta} \quad \bar{v}_{10} = \bar{v}_1 - \bar{V}$

$$\tan^2 \theta_1 = \frac{1 - \cos^2 \theta}{x^2 + 2x \cos \theta + \cos^2 \theta}$$

$$x^2 \tan^2 \theta_1 + 2x \cos \theta \tan^2 \theta_1 + \cos^2 \theta \tan^2 \theta_1 = 1 - \cos^2 \theta$$

$$(\sec^2 \theta_1) \cos^2 \theta + (2x \tan^2 \theta_1) \cos \theta + (x^2 \tan^2 \theta_1 - 1) = 0$$

$$\cos \theta = \frac{-2x \tan^2 \theta_1 \pm \sqrt{4x^2 \tan^4 \theta_1 - 4 \sec^2 \theta_1 (x^2 \tan^2 \theta_1 - 1)}}{2 \sec^2 \theta_1}$$

$$\cos \theta = -x \sin^2 \theta_1 \pm \sqrt{x^2 \sin^4 \theta_1 - x^2 \sin^2 \theta_1 + \cos^2 \theta_1}$$

$$\cos \theta = -x \sin^2 \theta_1 \pm \sqrt{-\cos^2 \theta_1 x^2 \sin^2 \theta_1 + \cos^2 \theta_1}$$

$$\cos \theta = -x \sin^2 \theta_1 \pm \cos \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1}$$

For $V < v_{10}$, $\cos \theta = -x \sin^2 \theta_1 + \cos \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1}$

$$= \frac{1}{2} \sin \theta d\theta = \frac{1}{2} \left[\frac{2x \sin \theta_1 \cos \theta_1 + \sin \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1} + \cos \theta_1 (2x \sin \theta_1 \cos \theta_1)}{2 \sqrt{1 - x^2 \sin^2 \theta_1}} \right] d\theta_1$$

$$= \frac{1}{2} \sin \theta_1 d\theta_1 \left(2x \cos \theta_1 + \sqrt{1 - x^2 \sin^2 \theta_1} + \frac{x^2 \cos^2 \theta_1}{\sqrt{1 - x^2 \sin^2 \theta_1}} \right)$$

$$\frac{1}{2} \sin \theta d\theta = \frac{1}{2} \sin \theta_1 d\theta_1 \left(2x \cos \theta_1 + \frac{1 + x^2 \cos 2\theta_1}{\sqrt{1 - x^2 \sin^2 \theta_1}} \right)$$

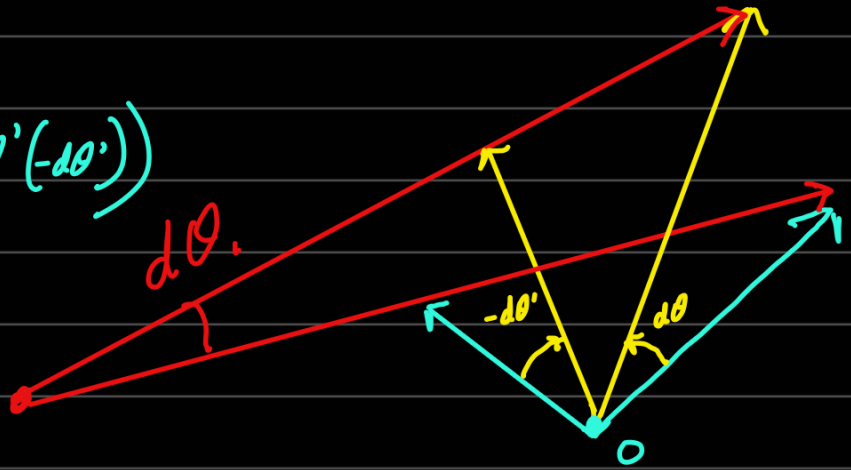
for $V > v_{10}$, $\frac{x > 1}{x > 1}$

$$\cos \theta = -x \sin^2 \theta_1 + \cos \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1}$$

$$\cos \theta' = -x \sin^2 \theta_1 - \cos \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1}$$

$$\frac{d\Omega_{total}}{4\pi} = \frac{d\Omega + d\Omega'}{4\pi}$$

$$= \frac{1}{2} (\sin \theta d\theta + \sin \theta' (-d\theta'))$$



Now

$$\sin \theta d\theta = d(x^2 \sin^2 \theta_1) + \sin \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1} d\theta_1 + \frac{x^2 \sin \theta_1 \cos^2 \theta_1}{\sqrt{1 - x^2 \sin^2 \theta_1}} d\theta_1$$

$$\sin \theta' d\theta' = d(x^2 \sin^2 \theta_1) - \sin \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1} d\theta_1 - \frac{x^2 \sin \theta_1 \cos^2 \theta_1}{\sqrt{1 - x^2 \sin^2 \theta_1}} d\theta_1$$

$$\therefore \frac{d\Omega_r}{4\pi} = \sin \theta_1 \sqrt{1 - x^2 \sin^2 \theta_1} d\theta_1 + \frac{x^2 \sin \theta_1 \cos^2 \theta_1}{\sqrt{1 - x^2 \sin^2 \theta_1}} d\theta_1$$

$$= \sin \theta_1 d\theta_1 \left(\frac{1 + x^2 \cos 2\theta_1}{\sqrt{1 - x^2 \sin^2 \theta_1}} \right)$$

2. Velocities in an elastic collision: Express the velocity of each particle after an elastic collision between a moving particle (say, m_1) and another at rest (say, m_2) in terms of their directions of motion in the laboratory system.

$$\bar{v}_{10} = \bar{v}_1 - \frac{m_1 \bar{v}_1 + m_2 \bar{v}_2}{m_1 + m_2}$$

$$\bar{v}_{10} = \frac{m_2 \bar{v}}{m_1 + m_2} \quad \bar{v} = \bar{v}_1 - \bar{v}_2$$

$$\bar{v}_{20} = \frac{-m_1 \bar{v}}{m_1 + m_2}$$

$$\bar{p}_{10} = m \bar{v}$$

$$\bar{p}_{20} = -m \bar{v}$$

$$m = \frac{m_1 m_2}{m_1 + m_2}$$

$$\therefore \bar{p}_1 = m\bar{v} + (\bar{p}_1 + \bar{p}_2) \frac{m_1}{m_1 + m_2} \quad \text{constant}$$

$$\bar{p}_2 = -m\bar{v} + (\bar{p}_1 + \bar{p}_2) \frac{m_2}{m_1 + m_2}$$

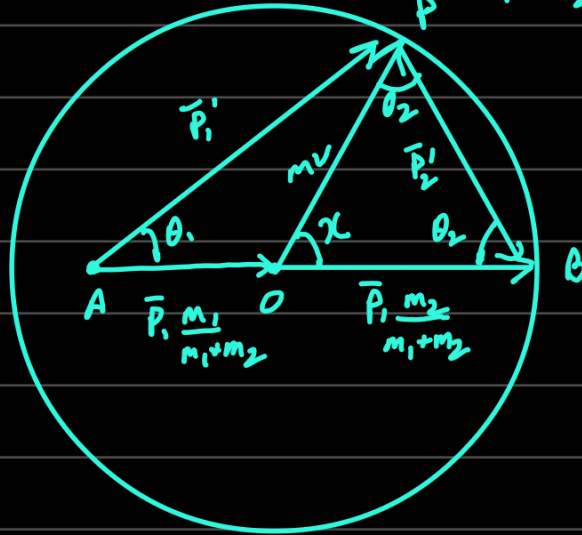
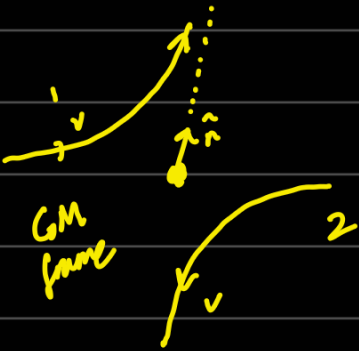
After collision, $\bar{p}'_1 = m v \hat{n} + \bar{p}_1 \frac{m_1}{m_1 + m_2}$

$$(\bar{v}_2 = 0, \bar{p}_2 = 0)$$

$$\bar{p}'_2 = -m v \hat{n} + \bar{p}_1 \frac{m_2}{m_1 + m_2}$$

$$\bar{v} = \bar{v}'_1 - \bar{v}'_2 \neq \bar{v}'_1$$

but $|\bar{v}| = |\bar{v}'_1|$



$$\tan \theta_1 = \frac{m v \sin \chi}{m v \cos \chi + \frac{m_1}{m_2} m v} = \frac{m_2 \sin \chi}{m_2 \cos \chi + m_1}$$

$$\theta_2 = \frac{1}{2} (\pi - \chi)$$

Now, $p_1'^2 = m^2 v^2 + \frac{p_1^2 m_1^2}{(m_1 + m_2)^2} + 2 m v p_1 \frac{m_1}{m_1 + m_2} \cos \chi$ $(\bar{p}'_1 = m\bar{v} + \bar{p}_1 \frac{m_1}{m_1 + m_2})$

$$m_1^2 v_1'^2 = \frac{m_1^2 m_2^2}{(m_1 + m_2)^2} v^2 + \frac{m_1^4 v_1^2}{(m_1 + m_2)^2} + 2 \frac{m_1^3 m_2}{(m_1 + m_2)^2} v_1^2 \cos \chi$$

$$v_1'^2 = \frac{m_2^2 v_1^2}{(m_1 + m_2)^2} + \frac{m_1^2 v_1^2}{(m_1 + m_2)^2} + \frac{2 m_1 m_2 v_1^2 \cos \chi}{(m_1 + m_2)^2}$$

$$\Rightarrow v_1' = \frac{v_1}{(m_1 + m_2)} \sqrt{m_1^2 + m_2^2 + 2 m_1 m_2 \cos \chi}$$

Similarly,
$$P_2'^2 = p_1^2 \frac{m_2^2}{(m_1+m_2)^2} + m_2^2 v^2 - 2 p_1 \frac{m_2}{m_1+m_2} m v \cos \chi$$

$$m_2^2 v_2'^2 = \frac{m_1^2 m_2^2 v_1^2}{(m_1+m_2)^2} + \frac{m_1^2 m_2^2 v_1^2}{(m_1+m_2)^2} - 2 \frac{m_1^2 m_2^2}{(m_1+m_2)^2} v_1^2 \cos \chi$$

$$v_2'^2 = \frac{2 m_1^2 v_1^2 - 2 m_1^2 v_1^2 \cos \chi}{(m_1+m_2)^2} = \frac{4 m_1^2 v_1^2 \sin^2(\chi/2)}{(m_1+m_2)^2}$$

$$\Rightarrow v_2' = \frac{2 m_1 v_1 \sin \chi/2}{m_1+m_2}$$

Express the velocity of each particle after a collision between a moving particle (m_1) and another at rest (m_2) in terms of their directions of motion in the L system.

SOLUTION. From Fig. 16 we have $p_2' = 2OB \cos \theta_2$ or $v_2' = 2v(m_1/m_2) \cos \theta_2$. The momentum $p_1' = AC$ is given by $OC^2 = AO^2 + p_1'^2 - 2AO \cdot p_1' \cos \theta_1$ or

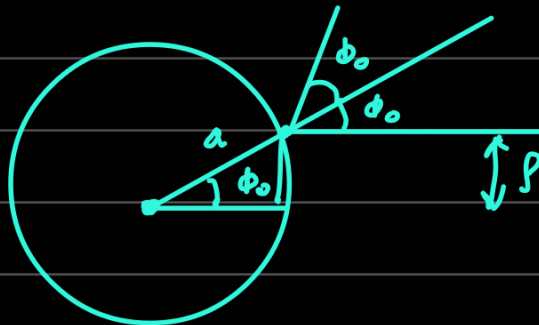
$$\left(\frac{v_1'}{v}\right)^2 - \frac{2m_1 v_1'}{m_2 v} \cos \theta_1 + \frac{m_1 - m_2}{m_1 + m_2} = 0.$$

Hence

$$\frac{v_1'}{v} = \frac{m_1}{m_1 + m_2} \cos \theta_1 \pm \frac{1}{m_1 + m_2} \sqrt{(m_2^2 - m_1^2 \sin^2 \theta_1)};$$

for $m_1 > m_2$ the radical may have either sign, but for $m_2 > m_1$ it must be taken positive.

3. Scattering by a rigid sphere: Consider scattering by a perfectly rigid sphere of radius a , i.e. when the interaction is such that $U(r) = \infty$ for $r < a$ and $U(r) = 0$ for $r > a$. Determine the effective cross-section for the scattering of particles by the rigid sphere. What is the total cross-section?



$$\chi = \pi - 2\phi_0$$

$$b = a \sin \phi_0 = a \sin \frac{\chi}{2}$$

$$d\sigma = \frac{b}{\sin \chi} \frac{db}{d\chi} d\chi$$

$$d\sigma = \frac{a \sin \chi/2}{\sin \chi} \frac{a}{2} \sin \frac{\chi}{2} d\chi$$

$$d\sigma = \frac{a^2}{4} d\chi$$

$$\Rightarrow \sigma = \pi a^2$$

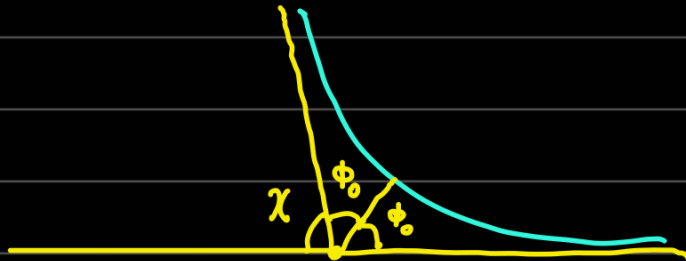
4. Rutherford scattering in the centre-of-mass frame: Consider scattering by the repulsive potential $U(r) = \alpha/r$ with $\alpha > 0$.

(a) Find the effective cross-section for scattering.

(b) What is the total cross-section? What does the result imply?

Note: This corresponds to the famous experiment by Rutherford wherein α -particles were scattered by atoms in a gold foil.

a)



Angle b/w asymptotes of hyperbola $= 2 \tan^{-1} \frac{b}{a} = 2 \phi_0$

$$\text{Also, } e = \sqrt{1 + \frac{2El^2}{m\alpha^2}} = \sqrt{1 + \frac{b^2}{a^2}}$$

input parameter $\uparrow$

$$\Rightarrow \frac{b}{a} = \sqrt{\frac{2El^2}{m\alpha^2}} \quad \text{and } E = \frac{mv^2}{2}, \quad l = mv\rho$$

$$\frac{b}{a} = \sqrt{\frac{mv^2 m^2 v^2 \rho^2}{m\alpha^2}} = \frac{mv^2 \rho}{\alpha}$$

$$\therefore \frac{mv^2 \rho}{\alpha} = \tan \phi_0 = \tan \left(\frac{1}{2} (\pi - \chi) \right) = \cot \frac{\chi}{2}$$

$$\Rightarrow \rho = \frac{\alpha}{mv^2} \cot \frac{\chi}{2}$$

$$\left| \frac{d\rho}{d\chi} \right| = \frac{\alpha}{2mv^2} \frac{1}{\sin^2 \frac{\chi}{2}}$$

Now, $d\sigma = \frac{\rho}{\sin \chi} \left| \frac{d\rho}{d\chi} \right| d\chi \rightarrow \text{solid angle}$

$$= \frac{1}{2 \sin \frac{\chi}{2} \cancel{\cos \frac{\chi}{2}}} \frac{\alpha}{mv^2} \frac{\cancel{\cos \frac{\chi}{2}}}{\sin \frac{\chi}{2}} \frac{\alpha}{2mv^2} \frac{1}{\sin^2 \frac{\chi}{2}} d\chi$$

$$d\sigma = \frac{\alpha^2}{4m^2 v^4 \sin^4 \frac{\chi}{2}} d\chi$$

b)

$$d\sigma = \frac{\alpha^2}{4m^2 v^4 \sin^4 \chi/2} \cdot \frac{4\pi \sin \chi \cos \chi}{2} d\chi$$

$$d\sigma = \frac{\pi \alpha^2}{m^2 v^4} \frac{\cos \chi/2}{\sin^3 \chi/2} d\chi$$

$$\sigma \rightarrow \infty \quad \text{because } \frac{1}{r} \text{ is a long range potential.}$$

5. Scattering cross-section in the laboratory frame: Determine the cross-section in the laboratory frame in the cases of scattering by a hard sphere and Rutherford scattering.

Hard Sphere

$$d\sigma = \frac{1}{4} a^2 d\phi = \frac{\pi a^2}{2} \sin \chi d\chi = \frac{\pi a^2}{2} d(\cos \chi)$$

$$\tan \theta_1 = \frac{m_2 \sin \chi}{m_2 \cos \chi + m_1} = \frac{\sin \chi}{\cos \chi + \mu}, \quad \mu = \frac{m_1}{m_2}$$

$$\tan^2 \theta_1 = \frac{1 - \cos^2 \chi}{\cos^2 \chi + \mu^2 + 2\mu \cos \chi}$$

$$\sec^2 \theta_1 \cos^2 \chi + \mu^2 \tan^2 \theta_1 + 2\mu \cos \chi \tan^2 \theta_1 - 1 = 0$$

$$\cos \chi = \frac{-2\mu \tan^2 \theta_1 \pm \sqrt{4\mu^2 \tan^4 \theta_1 - 4\mu^2 \tan^2 \theta_1 \sec^2 \theta_1 + 4\sec^2 \theta_1}}{2\sec^2 \theta_1}$$

$$= -\mu \sin^2 \theta_1 \pm \sqrt{\mu^2 \sin^4 \theta_1 - \mu^2 \sin^2 \theta_1 + \cos^2 \theta_1}$$

$$\cos \chi = -\mu \sin^2 \theta_1 \pm \cos \theta_1 \sqrt{1 - \mu^2 \sin^2 \theta_1}$$

Rutherford Scattering

$$d\sigma = \frac{\alpha^2}{4m^2v^4 \sin^4 \chi/2} \quad d\sigma = \frac{4 \cos^4 \chi/2 \alpha^2}{m^2v^4 \sin^4 \chi} d\sigma$$

$$\cos \chi = -\mu \sin^2 \theta_1 \pm \cos \theta_1 \sqrt{1 - \mu^2 \sin^2 \theta_1}$$